

Penetration Test Report

Security Assessment and Vulnerability Analysis

Target Scope:	192.168.56.101/32
Scan Date:	April 27, 2026 at 14:28:32 UTC
Port Range:	1-200
Overall Risk Rating:	CRITICAL

1. Executive Summary

This penetration test was conducted on April 27, 2026, against the target system at IP address 192.168.56.101. The assessment involved comprehensive reconnaissance and vulnerability analysis of network services running on ports 1 through 200. The objective was to identify security vulnerabilities that could be exploited by malicious actors to compromise the confidentiality, integrity, or availability of the system.

The assessment revealed a **CRITICAL** overall risk level. A total of eight (8) vulnerabilities were identified, including two (2) critical-severity issues, three (3) high-severity issues, two (2) medium-severity issues, and one (1) low-severity issue. The most critical findings include a backdoor in vsftpd 2.3.4 (CVE-2011-2523) that allows remote code execution, and a command injection vulnerability in Samba 3.0.20 (CVE-2007-2447) that also enables remote code execution. These vulnerabilities could allow an attacker to gain complete control of the system without authentication.

Immediate remediation is required for the critical vulnerabilities. The organization should prioritize upgrading outdated software, disabling insecure protocols such as Telnet and FTP, and implementing strong cryptographic standards across all services. The presence of multiple end-of-life software components indicates a significant patch management deficit that must be addressed urgently to reduce the attack surface and prevent potential compromise.

2. Scope and Methodology

The scope of this penetration test was limited to the single IP address 192.168.56.101 with a subnet mask of /32. The assessment focused on TCP ports in the range of 1 through 200. Network reconnaissance and vulnerability identification were performed using industry-standard tools including Nmap (version 7.99) for port scanning and service enumeration, Gobuster for web directory discovery, cURL for HTTP response analysis, and WHOIS for domain information gathering.

This assessment was conducted as a non-intrusive reconnaissance and vulnerability identification exercise. No exploitation attempts were performed. Specifically, the assessment did NOT include: active exploitation of identified vulnerabilities, authentication or login attempts using default or weak credentials, modification of files or system configurations, deployment of shellcode or payloads, denial-of-service attacks, or social engineering activities. The findings are based solely on service version identification, configuration analysis, and comparison against known vulnerability databases.

3. Risk Matrix

The following risk matrix summarizes the distribution of vulnerabilities by severity level:

Severity Level	Count
Critical	2
High	3
Medium	2
Low	1

4. Overall Risk Rating

Risk Level: **CRITICAL**

The overall risk rating for this system is CRITICAL due to the presence of two vulnerabilities with critical severity ratings. Both critical vulnerabilities allow for unauthenticated remote code execution, which represents the highest possible threat to system security. An attacker could leverage these vulnerabilities to gain complete administrative control over the target system, install malware, exfiltrate sensitive data, pivot to other systems on the network, or cause system disruption. The presence of multiple high-severity vulnerabilities further compounds the risk, as they provide additional attack vectors including cleartext credential exposure, end-of-life software with known exploits, and weak cryptographic implementations. Immediate action is required to address these critical security deficiencies.

5. Detailed Findings

ID	Title	Severity	Evidence	CVE	Remediation
VULN-001	vsftpd 2.3.4 Backdoor - Remote Code Execution	Critical	nmap detected vsftpd 2.3.4 running on port 21/tcp with anonymous FTP enabled (FTP code 230)	CVE-2011-2523	Upgrade to vsftpd 2.3.5 or later. Remove the compromised binary. Disable anonymous FTP access.
VULN-002	OpenSSH 4.7p1 - Weak Cryptography	Low	nmap detected OpenSSH 4.7p1 on port 22/tcp with 1024-bit DSA host key	CVE-2008-5161	Upgrade to OpenSSH 8.0 or later. Disable weak ciphers and key exchange algorithms. Replace DSA keys with Ed25519 or RSA 4096-bit keys.
VULN-003	Telnet - Cleartext Protocol	High	nmap detected Linux telnetd running on port 23/tcp	No CVE	Disable telnet service immediately. Use SSH for remote access instead. Block port 23 at the firewall level.
VULN-004	SMTP - SSLv2 Enabled with Weak Ciphers	Medium	nmap script detected SSLv2 enabled on port 25/tcp with weak ciphers including SSL2_RC4_128_EXPORT40_WITH_MD5. SSL certificate expired 2010-04-16.	No CVE	Disable SSLv2 and SSLv3. Configure TLS 1.2 or higher only. Use strong cipher suites. Renew SSL certificate.
VULN-005				No CVE	

ID	Title	Severity	Evidence	CVE	Remediation
	ISC BIND 9.4.2 - End-of-Life DNS Server	High	nmap detected ISC BIND 9.4.2 on port 53/tcp via dns-nsid script		Upgrade to ISC BIND 9.18 or later. Consider using a maintained DNS server like PowerDNS or Unbound.
VULN-006	Apache 2.2.8 - ByteRange DoS and Information Disclosure	High	nmap detected Apache httpd 2.2.8 on port 80/tcp. Gobuster found /phpinfo.php (48KB) and /dav directory. X-Powered-By header reveals PHP/5.2.4-2ubuntu5.10.	CVE-2011-3192	Upgrade to Apache 2.4.x. Remove phpinfo.php. Disable WebDAV if not required. Update PHP to supported version.
VULN-007	RPC Service - Information Disclosure	Medium	nmap rpcinfo script enumerated 10 RPC services including NFS v2/3/4 on ports 2049 TCP/UDP	No CVE	Restrict rpcbind to localhost only if not required for legitimate clients. Use firewall rules to block external access to port 111 and related RPC services.
VULN-008	Samba 3.0.20 - Remote Code Execution via Username Map Script	Critical	nmap detected Samba 3.0.20-Debian on ports 139 and 445. SMB security scan shows guest account enabled, message signing disabled, system time reveals hostname metasploitable.localdomain.	CVE-2007-2447	Upgrade Samba to >= 3.0.25 or current stable (4.x). Disable username map script. Enable SMB signing.

ID	Title	Severity	Evidence	CVE	Remediation
					Disable guest access.

Appendix A: Raw Tool Outputs

Nmap Scan Results

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE nmaprun>
<?xml-stylesheet href="file:///C:/Program Files (x86)/Nmap/nmap.xsl" type="text/xsl"?>
<!-- Nmap 7.99 scan initiated Mon Apr 27 14:28:32 2026 as: nmap -sV -O -sC --open -p
1-200 -oX - 192.168.56.101 -->
<nmaprun scanner="nmap" args="nmap -sV -O -sC --open -p 1-200 -oX - 192.168.56.101"
start="1777314512" startstr="Mon Apr 27 14:28:32 2026" version="7.99"
xmloutputversion="1.05">
<scaninfo type="syn" protocol="tcp" numservices="200" services="1-200"/>
<verbose level="0"/>
<debugging level="0"/>
<hosthint><status state="up" reason="arp-response" reason_ttl="0"/>
<address addr="192.168.56.101" addrtype="ipv4"/>
<address addr="08:00:27:B0:8F:13" addrtype="mac" vendor="Oracle VirtualBox virtual NIC"/
>
<hostnames>
</hostnames>
</hosthint>
<host starttime="1777314514" endtime="1777314551"><status state="up" reason="arp-
response" reason_ttl="0"/>
<address addr="192.168.56.101" addrtype="ipv4"/>
<address addr="08:00:27:B0:8F:13" addrtype="mac" vendor="Oracle VirtualBox virtual NIC"/
>
<hostnames>
</hostnames>
<ports><extraports state="closed" count="192">
<extrareasons reason="reset" count="192" proto="tcp"
ports="1-20,24,26-52,54-79,81-110,112-138,140-200"/>
</extraports>
<port protocol="tcp" portid="21"><state state="open" reason="syn-ack" reason_ttl="64"/
><service name="ftp" product="vsftpd" version="2.3.4" ostype="Unix" method="probed"
conf="10"><cpe>cpe:/a:vsftpd:vsftpd:2.3.4</cpe></service><script id="ftp-anon"
output="Anonymous FTP login allowed (FTP code 230)"/><script id="ftp-syst" output="
STAT:
FTP server status:
  Connected to 192.168.56.1
  Logged in as ftp
  TYPE: ASCII
  No session bandwidth limit
  Session timeout in seconds is 300
  Control connection is plain text
  Data connections will be plain text
  vsFTPD 2.3.4 - secure, fast, stable
End of status"><elem key="STAT">
FTP server status:
  Connected to 192.168.56.1
  Logged in as ftp
  TYPE: ASCII
  No session bandwidth limit
  Session timeout in seconds is 300
```

```

Control connection is plain text
Data connections will be plain text
vsFTPD 2.3.4 - secure, fast, stable
End of status</elem>
</script></port>
<port protocol="tcp" portid="22"><state state="open" reason="syn-ack" reason_ttl="64"/>
<service name="ssh" product="OpenSSH" version="4.7p1 Debian 8ubuntu1"
extrainfo="protocol 2.0" ostype="Linux" method="probed" conf="10"><cpe>cpe:/
a:openbsd:openssh:4.7p1</cpe><cpe>cpe:/o:linux:linux_kernel</cpe></service><script
id="ssh-hostkey" output="
1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)"><table>
<elem key="fingerprint">600fcfe1c05f6a74d69024fac4d56ccd</elem>
<elem key="key">AAAAB3NzaC1kc3MAAACBALz4hsc8a2SrQ4nlW960qV8xwBG0JC+jI7fWxm5METIJH4tKr/
xUTwsTYEYnaZLzcOiy21D3ZvOwYb6AA3765zdgCd2Tgand7F0YD5UtXG7b7fbz99chReivL0SIWEG/
E96Ai+pqYMP2WD5Ka0JwSIXSUAjnuU5oWmY5x85sBw+XDAAAQDFkMpmDFQTF+oRqaoSNVU7Z+hjSwAAAIBCQxNK
zi1TyP+QJIFa3M0oLqCVWIOWe/ArtXrzpBOJ/
dt0hTJXCeYisKqcdwdtyIn8OUCOyrIjQNuA2QW217oQ6wXpbFh+5AQm8H13b6C6o8lX3Ptw+Y4dp0lzfWHwZ/
jzHwtuaDQaok7u1f9711EazeJLqfiWrAzoklqSWyDQJAAAAIA1lAD3xWYkeIeHv/
R3P9i+XaoI7imFkMuYXCDTq843YU6Td+0mWp1lCqAWUV/
CQamGgQLtYy5S0ueoks01MoKdOMMhKVwqdr08nvCBdNKjIEd3gH6oBk/YRnjzxleAYBsvCmM4a0jmhZ0oNiRWlc/
F+bkUeFkrBx/D2fdfZmhrGg==</elem>
<elem key="bits">1024</elem>
<elem key="type">ssh-dss</elem>
</table>
</script></port>
<port protocol="tcp" portid="23"><state state="open" reason="syn-ack" reason_ttl="64"/>
<service name="telnet" product="Linux telnetd" ostype="Linux" method="probed"
conf="10"><cpe>cpe:/o:linux:linux_kernel</cpe></service></port>
<port protocol="tcp" portid="25"><state state="open" reason="syn-ack" reason_ttl="64"/>
<service name="smtp" product="Postfix smtpd" hostname="metasploitable.localdomain"
method="probed" conf="10"><cpe>cpe:/a:postfix:postfix</cpe></service><script id="sslv2"
output="
SSLv2 supported
ciphers:
SSL2_RC4_128_EXPORT40_WITH_MD5
SSL2_RC4_128_WITH_MD5
SSL2_RC2_128_CBC_EXPORT40_WITH_MD5
SSL2_DES_64_CBC_WITH_MD5
SSL2_DES_192_EDE3_CBC_WITH_MD5
SSL2_RC2_128_CBC_WITH_MD5"><elem>SSLv2 supported</elem>
<table key="ciphers">
<elem>SSL2_RC4_128_EXPORT40_WITH_MD5</elem>
<elem>SSL2_RC4_128_WITH_MD5</elem>
<elem>SSL2_RC2_128_CBC_EXPORT40_WITH_MD5</elem>
<elem>SSL2_DES_64_CBC_WITH_MD5</elem>
<elem>SSL2_DES_192_EDE3_CBC_WITH_MD5</elem>
<elem>SSL2_RC2_128_CBC_WITH_MD5</elem>
</table>
</script><script id="smtp-commands" output="metasploitable.localdomain, PIPELINING, SIZE
10240000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES, 8BITMIME, DSN"/><script id="ssl-
date" output="2026-04-27T18:27:05+00:00; -2m05s from scanner time."><elem
key="delta">-125.0</elem>
<elem key="date">2026-04-27T18:27:05+00:00</elem>
</script><script id="ssl-cert" output="Subject: commonName=ubuntu804-base.localdomain/
organizationName=OCOSA/stateOrProvinceName=There is no such thing outside US/
countryName=XX
Not valid before: 2010-03-17T14:07:45
Not valid after: 2010-04-16T14:07:45"></script></port>
<port protocol="tcp" portid="53"><state state="open" reason="syn-ack" reason_ttl="64"/

```



```

<>service name="domain" product="ISC BIND" version="9.4.2" method="probed"
conf="10"><cpe>cpe:/a:isc:bind:9.4.2</cpe></service><script id="dns-nsid" output="
  bind.version: 9.4.2"><elem key="bind.version">9.4.2</elem>
</script></port>
<port protocol="tcp" portid="80"><state state="open" reason="syn-ack" reason_ttl="64"/
><service name="http" product="Apache httpd" version="2.2.8" extrainfo="(Ubuntu) DAV/2"
method="probed" conf="10"><cpe>cpe:/a:apache:http_server:2.2.8</cpe></service><script
id="http-title" output="Metasploitable2 - Linux"><elem key="title">Metasploitable2 -
Linux</elem>
</script><script id="http-server-header" output="Apache/2.2.8 (Ubuntu) DAV/
2"><elem>Apache/2.2.8 (Ubuntu) DAV/2</elem>
</script></port>
<port protocol="tcp" portid="111"><state state="open" reason="syn-ack" reason_ttl="64"/
><service name="rpcbind" version="2" extrainfo="RPC #100000" method="probed" conf="10"/
><script id="rpcinfo" output="
  program version      port/proto  service
100000  2                111/tcp    rpcbind
100000  2                111/udp    rpcbind
100003  2,3,4            2049/tcp   nfs
100003  2,3,4            2049/udp   nfs
100005  1,2,3            39913/tcp  mountd
100005  1,2,3            44533/udp  mountd
100021  1,3,4            34088/udp  nlockmgr
100021  1,3,4            42238/tcp  nlockmgr
100024  1                34928/tcp  status
100024  1                50963/udp  status
"></script></port>
<port protocol="tcp" portid="139"><state state="open" reason="syn-ack" reason_ttl="64"/
><service name="netbios-ssn" product="Samba smbd" version="3.0.20-Debian"
extrainfo="workgroup: WORKGROUP" method="probed" conf="10"><cpe>cpe:/a:samba:samba</
cpe></service></port>
</ports>
<os><portused state="open" proto="tcp" portid="21"/>
<portused state="closed" proto="tcp" portid="1"/>
<portused state="closed" proto="udp" portid="39856"/>
<osmatch name="Linux 2.6.9 - 2.6.33" accuracy="100" line="63177">
<osclass type="general purpose" vendor="Linux" osfamily="Linux" osgen="2.6.X"
accuracy="100"><cpe>cpe:/o:linux:linux_kernel:2.6</cpe></osclass>
</osmatch>
</os>
<uptime seconds="3046" lastboot="Mon Apr 27 13:38:25 2026"/>
<distance value="1"/>
<tcpsequence index="203" difficulty="Good luck!"
values="2B555D06,2AC76F17,2B6E06B5,2AB11C43,2B66E710,2B835BCA"/>
<ipidsequence class="All zeros" values="0,0,0,0,0,0"/>
<tcptssequence class="100HZ" values="49BFA,49C04,49C0E,49C18,49C22,49C2C"/>
<hostscript><script id="nbstat" output="NetBIOS name: METASPLOITABLE, NetBIOS user:
<unknown>, NetBIOS MAC: <unknown> (unknown)"/><script id="smb-security-mode" output="
  account_used: guest
  authentication_level: user
  challenge_response: supported
  message_signing: disabled (dangerous, but default)"><elem key="account_used">guest</
elem>
<elem key="authentication_level">user</elem>
<elem key="challenge_response">supported</elem>
<elem key="message_signing">disabled</elem>
</script><script id="smb2-time" output="Protocol negotiation failed (SMB2)"/><script
id="clock-skew" output="mean: 1h17m54s, deviation: 2h18m33s, median: -2m05s"><elem
key="mean">4674</elem>
<elem key="stddev">8313</elem>

```

```

<elem key="count">3</elem>
<elem key="median">-125</elem>
</script><script id="smb-os-discovery" output="
  OS: Unix (Samba 3.0.20-Debian)
  Computer name: metasploitable
  NetBIOS computer name:
  Domain name: localdomain
  FQDN: metasploitable.localdomain
  System time: 2026-04-27T14:26:41-04:00
"><elem key="os">Unix</elem>
<elem key="lanmanager">Samba 3.0.20-Debian</elem>
<elem key="server"></elem>
<elem key="date">2026-04-27T14:26:41-04:00</elem>
<elem key="fqdn">metasploitable.localdomain</elem>
<elem key="domain_dns">localdomain</elem>
<elem key="workgroup">WORKGROUP\x00</elem>
</script></hostscript><times srtt="2922" rttvar="2015" to="100000"/>
</host>
<runstats><finished time="1777314551" timestr="Mon Apr 27 14:29:11 2026" summary="Nmap
done at Mon Apr 27 14:29:11 2026; 1 IP address (1 host up) scanned in 38.27 seconds"
elapsed="38.27" exit="success"/><hosts up="1" down="0" total="1"/>
</runstats>
</nmaprun>

```

Gobuster Web Directory Enumeration

.hta	(Status: 403) [Size: 291]
.htaccess	(Status: 403) [Size: 296]
.htpasswd	(Status: 403) [Size: 296]
cgi-bin/	(Status: 403) [Size: 295]
dav	(Status: 301) [Size: 319] [--> http://192.168.56.101/dav/]
index	(Status: 200) [Size: 891]
index.php	(Status: 200) [Size: 891]
phpMyAdmin	(Status: 301) [Size: 326] [--> http://192.168.56.101/phpMyAdmin/]
phpinfo	(Status: 200) [Size: 48017]
phpinfo.php	(Status: 200) [Size: 48029]
server-status	(Status: 403) [Size: 300]
test	(Status: 301) [Size: 320] [--> http://192.168.56.101/test/]
twiki	(Status: 301) [Size: 321] [--> http://192.168.56.101/twiki/]

cURL HTTP Response Headers

```

HTTP/1.1 200 OK
Date: Mon, 27 Apr 2026 18:27:18 GMT
Server: Apache/2.2.8 (Ubuntu) DAV/2
X-Powered-By: PHP/5.2.4-2ubuntu5.10
Content-Type: text/html

```

WHOIS Information

No WHOIS data available

Confidential - For Internal Use Only

This report contains sensitive security information and should be handled accordingly.

Report Generated: April 27, 2026